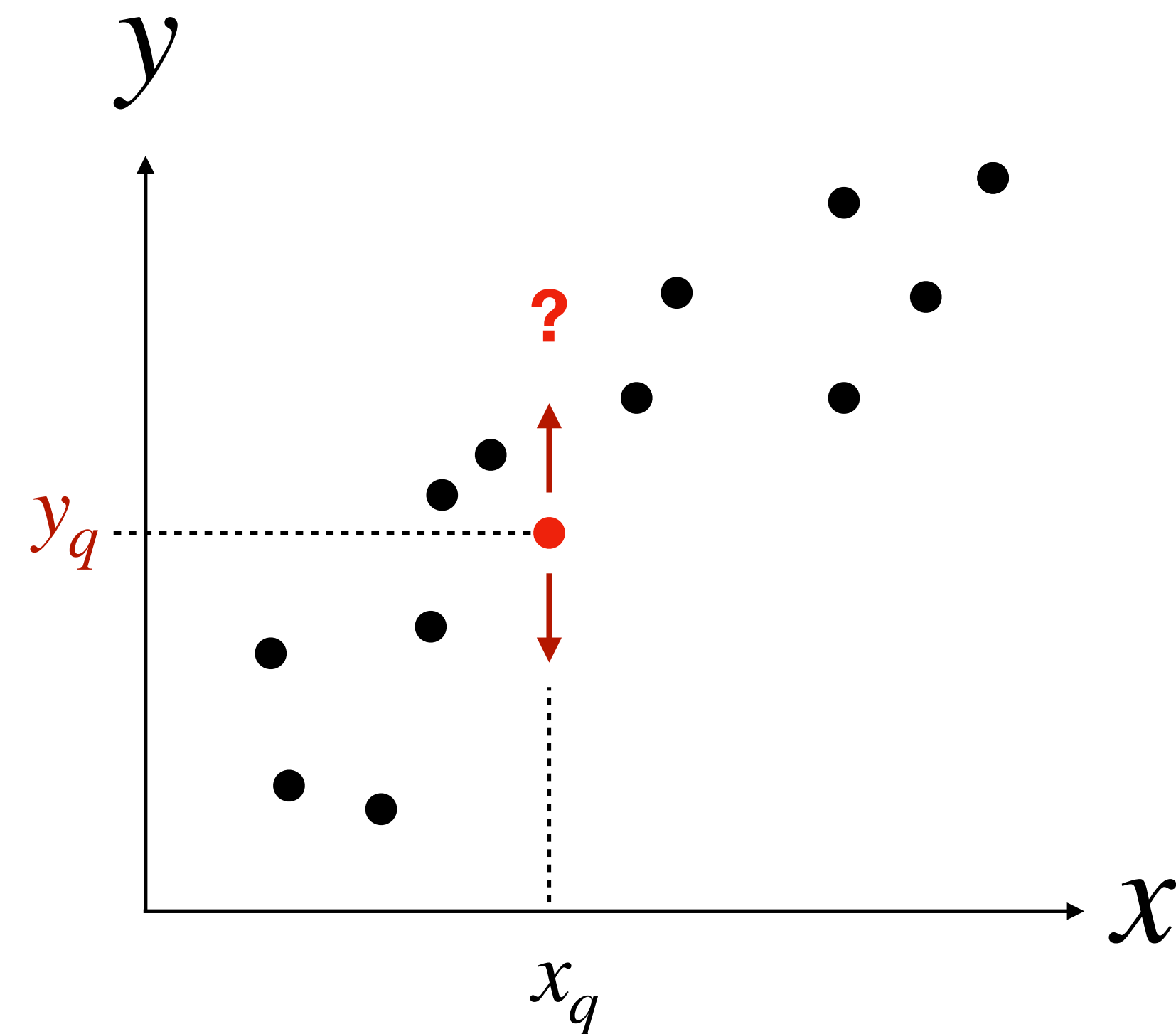
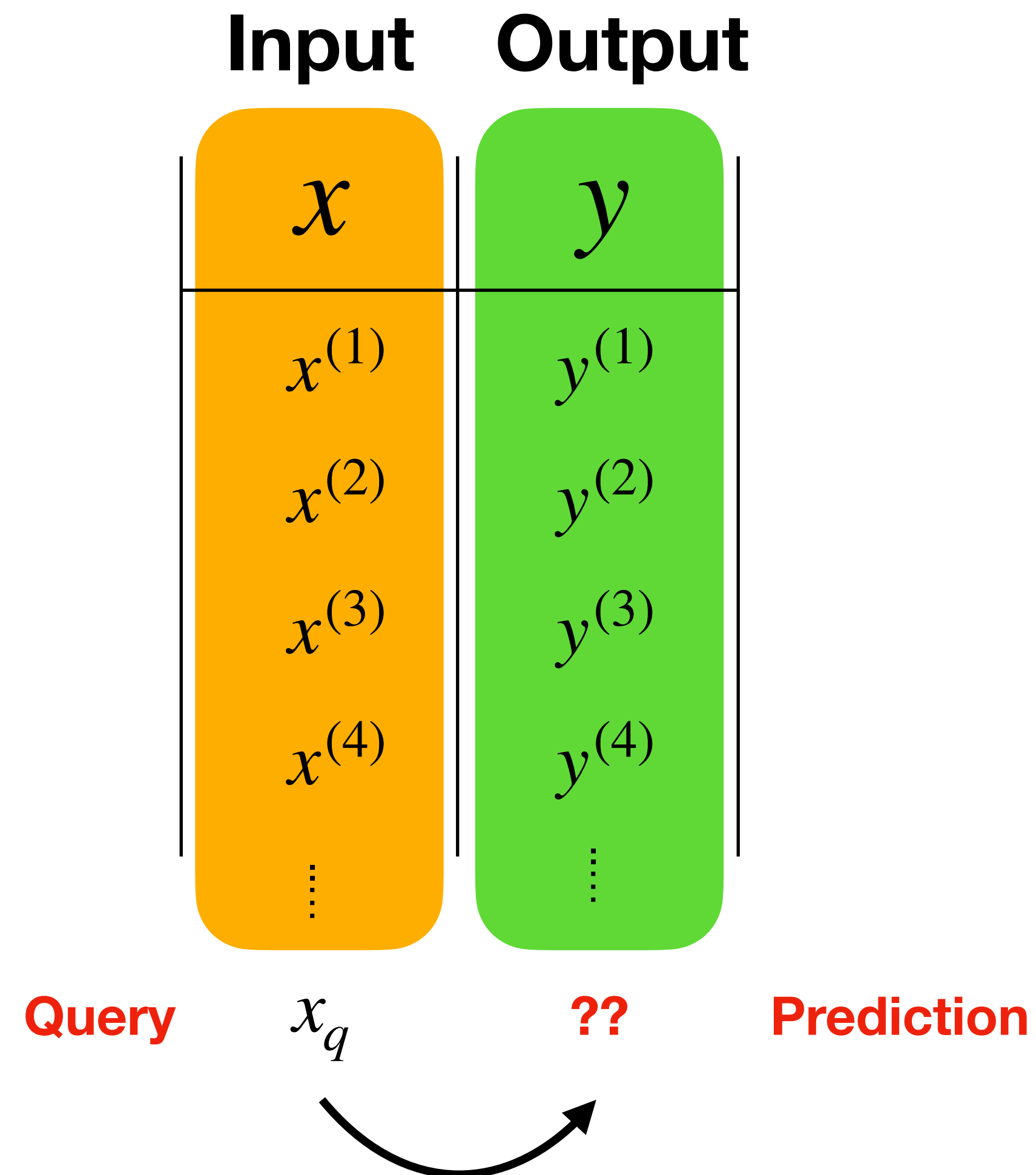


Logistic Regression

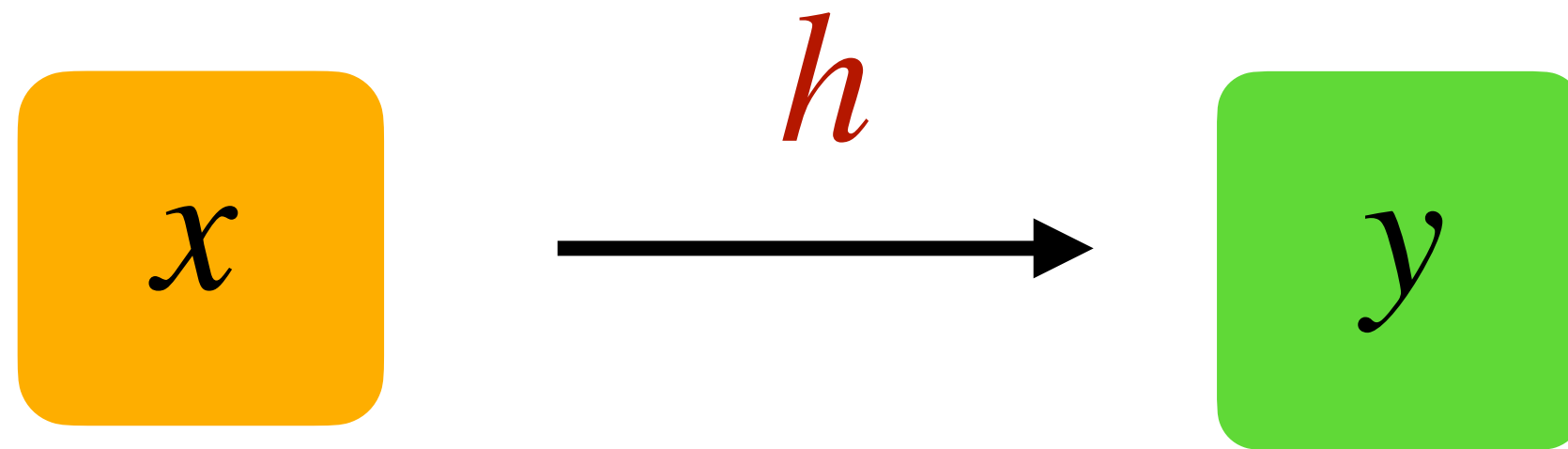
Prepared by: Joseph Bakarji

Given new input, what's the output?



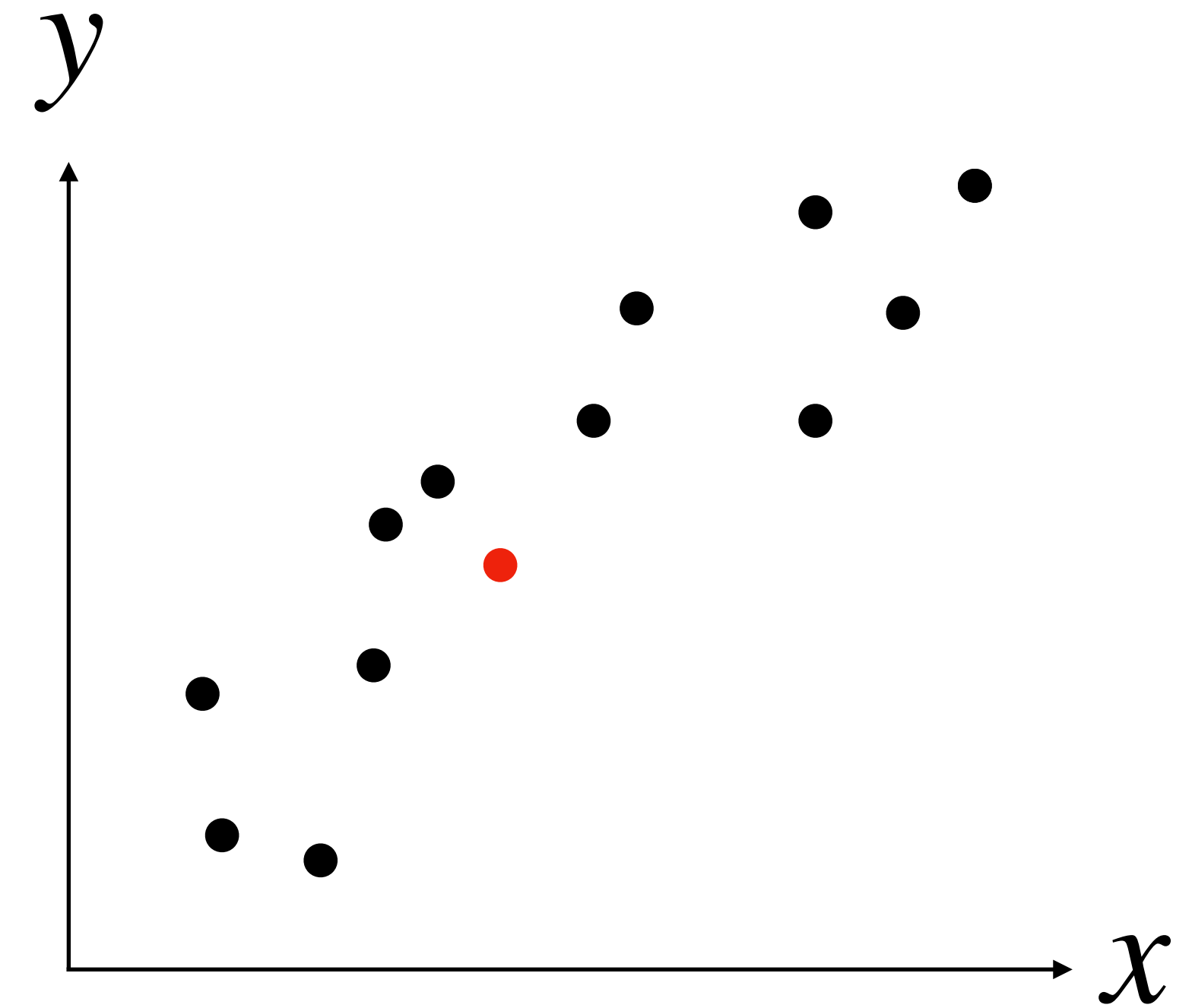
Given new input, what's the output?

Assuming $y \in \mathbb{R}$

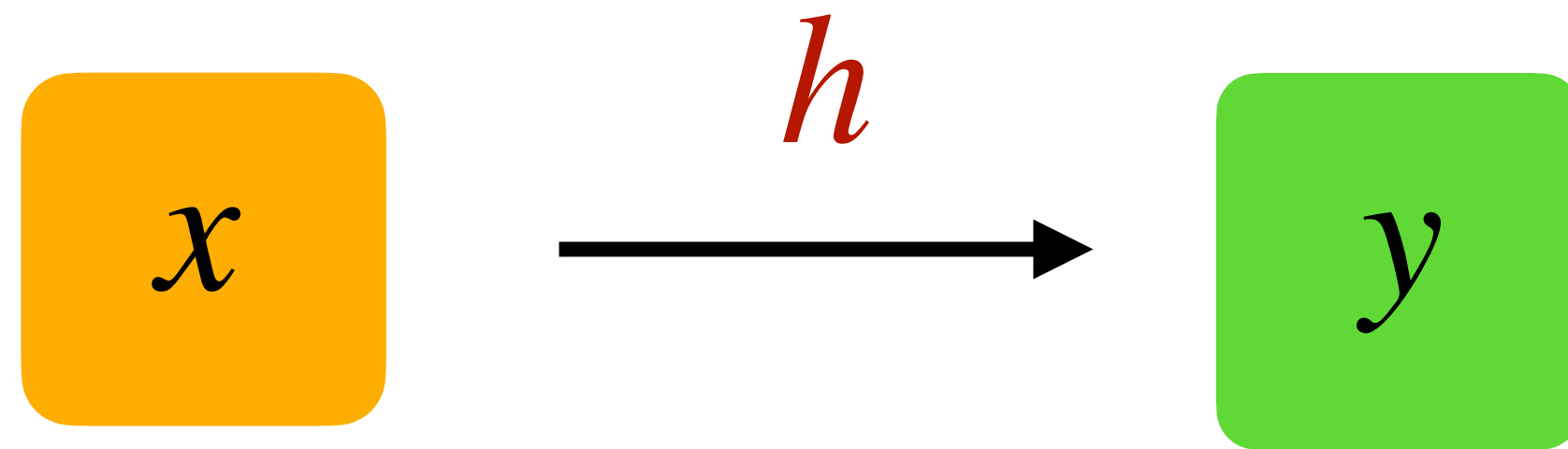


Given the data,
find a **function** h ,
that predicts y , given x

$$y = h(x)$$



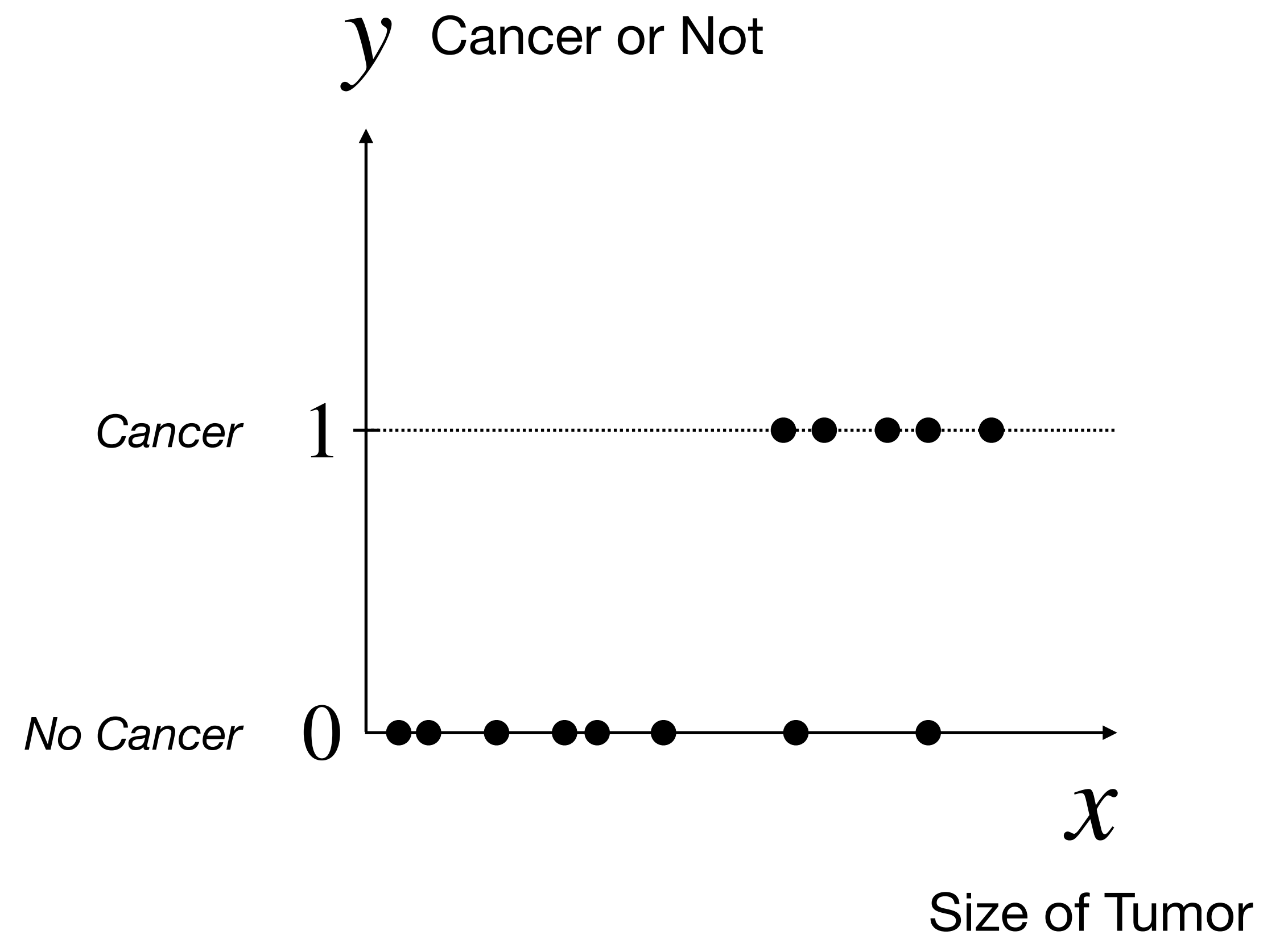
What if y is a label?



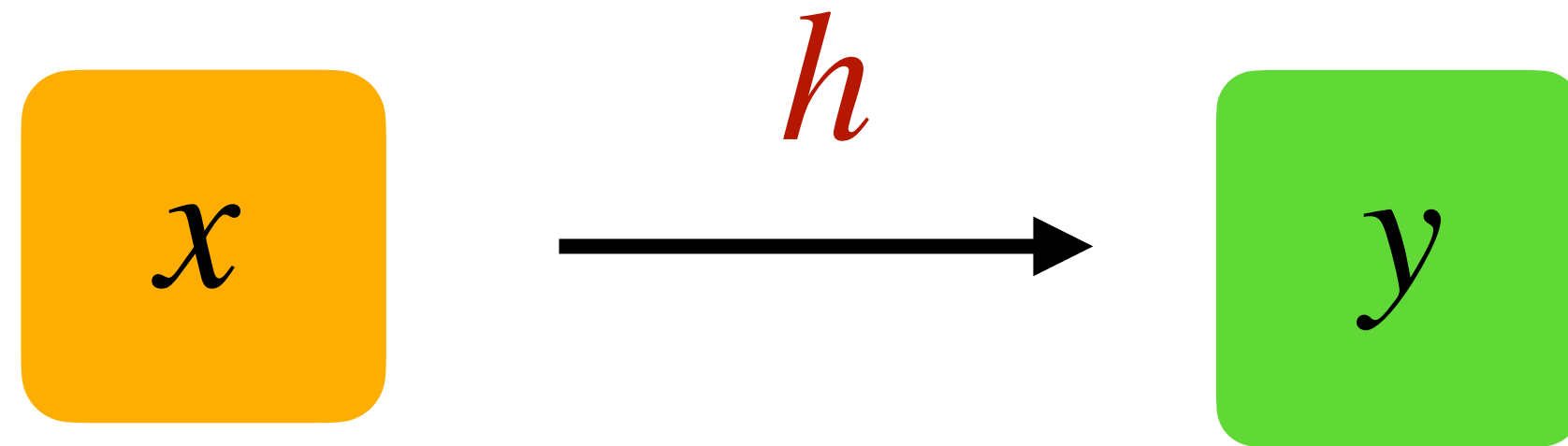
Given the data,
find a **function** h ,
that predicts y , given x

$$y = h(\mathbf{x})$$

$$y \in [0,1]$$



What if y is a label?

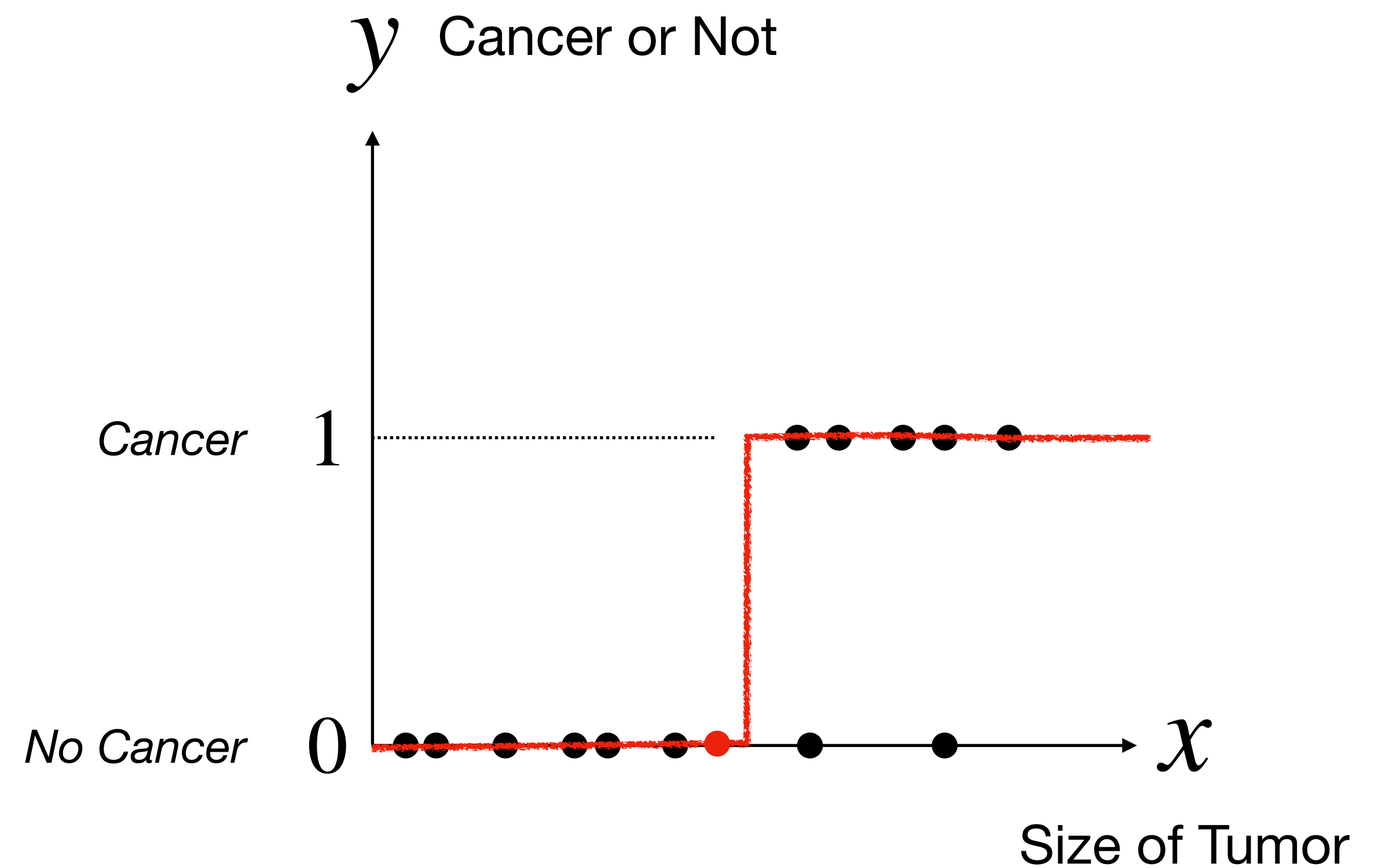


Given the data,
find a **function** h ,
that predicts y , given x

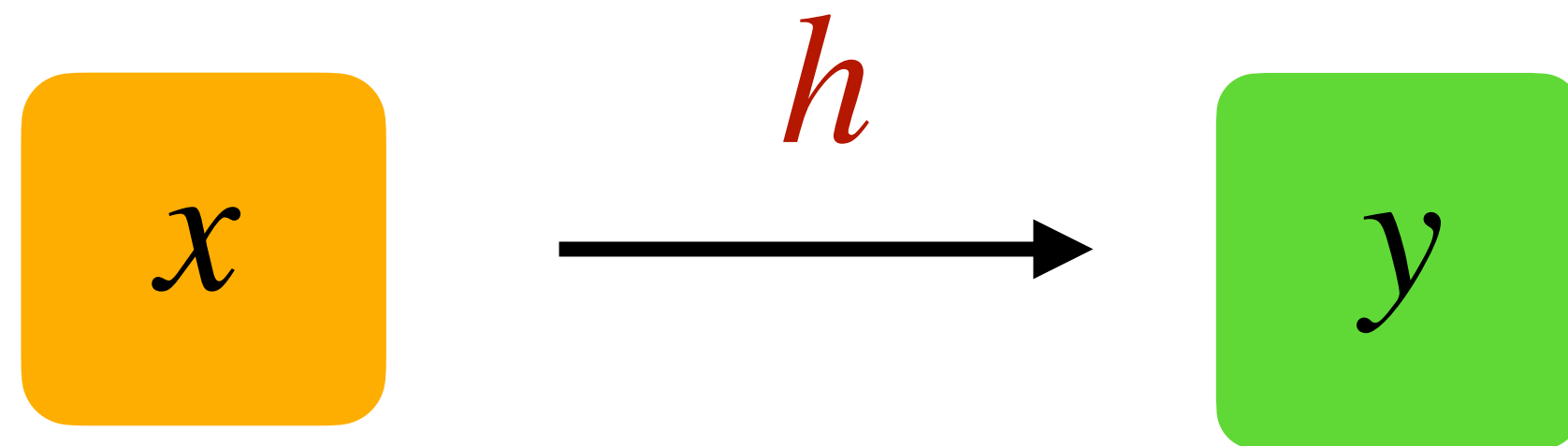
$$y = h(\mathbf{x})$$

$$y \in [0, 1]$$

A step function, or threshold



What if y is a label?

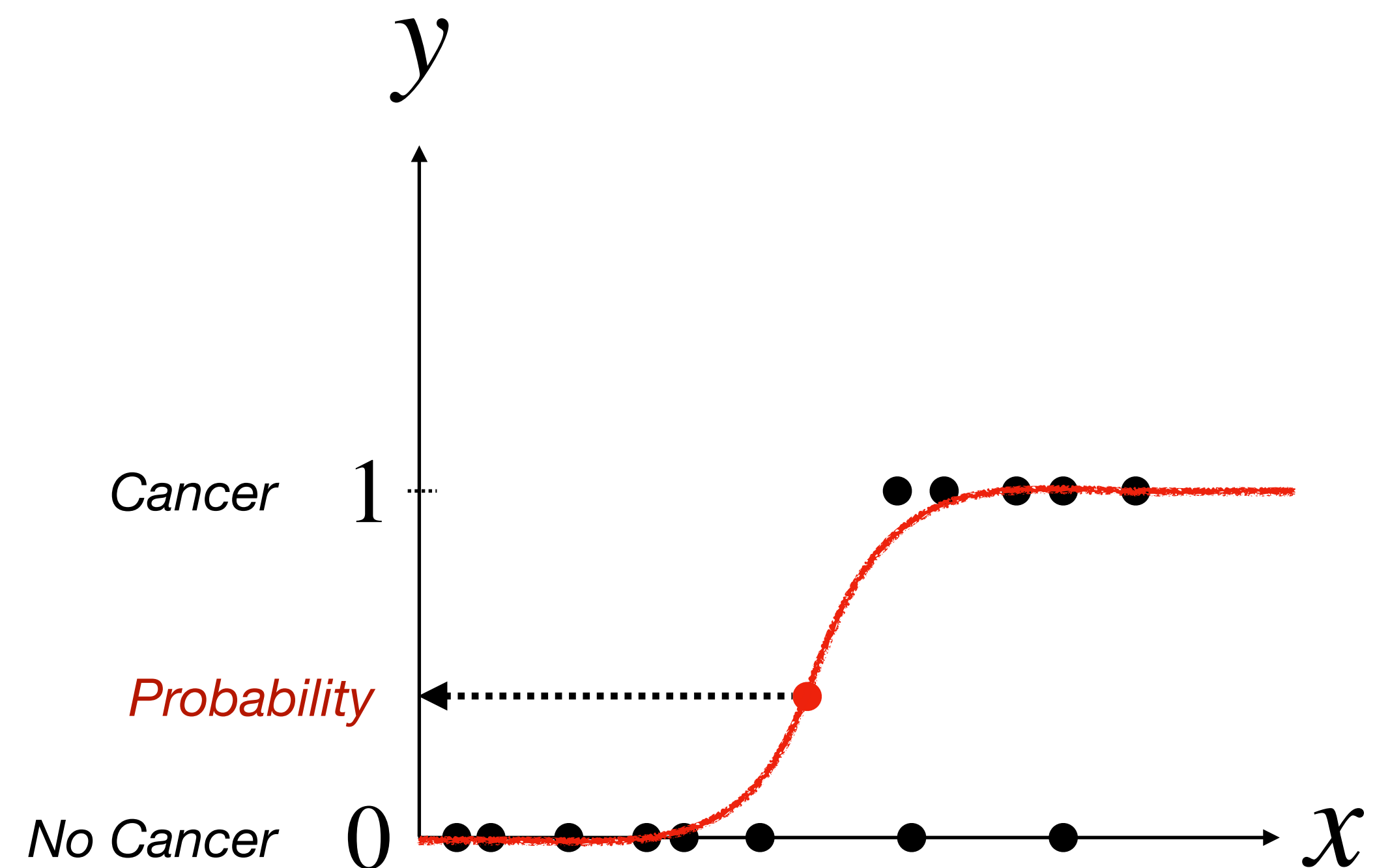


Given the data,
find a **function** h ,
that predicts y , given x

$$y = h(\mathbf{x})$$

$$y \in [0, 1]$$

A **smooth** function that returns
probability of occurrence

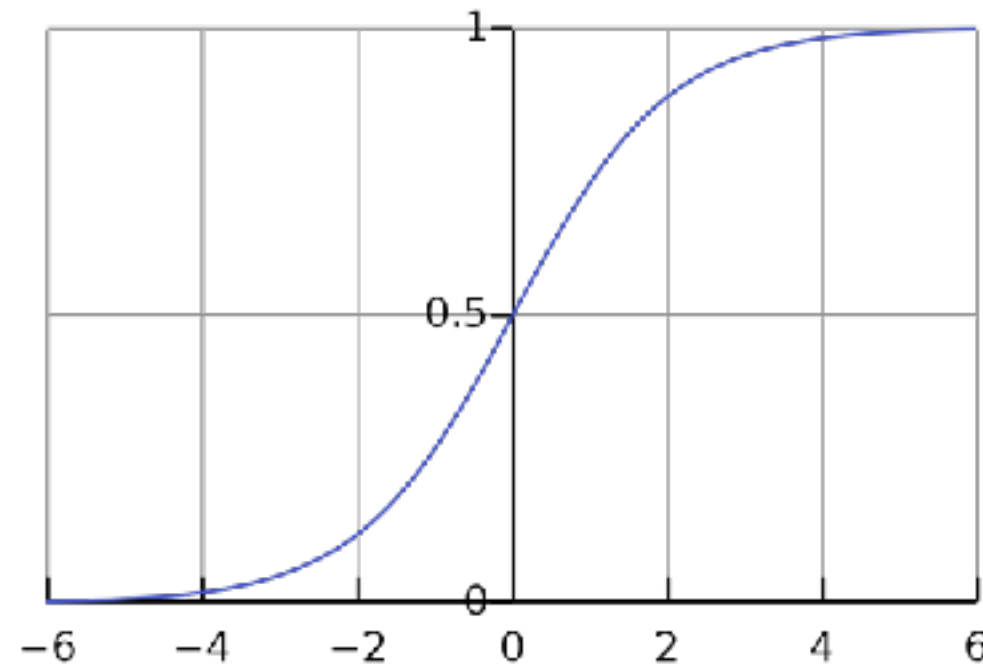


What if y is a label?

$$y = h_{\theta}(x) \quad \& \quad y \in [0,1]$$

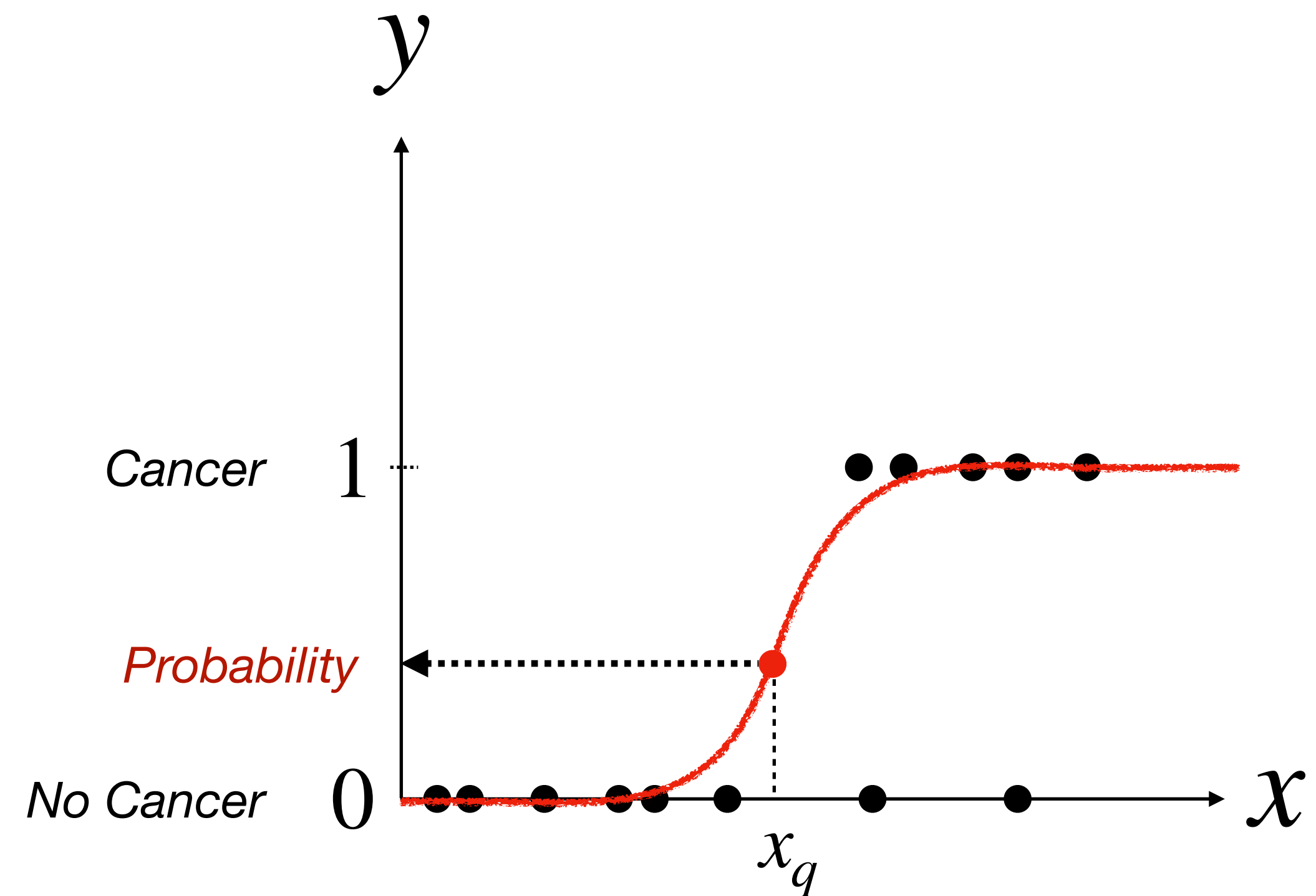
Logistic Function

$$y = \frac{1}{1 + e^{-x}}$$



A **smooth** function that returns **probability of occurrence**

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$



What if y is a label?

$$y = h_{\theta}(x) \quad \& \quad y \in [0,1]$$

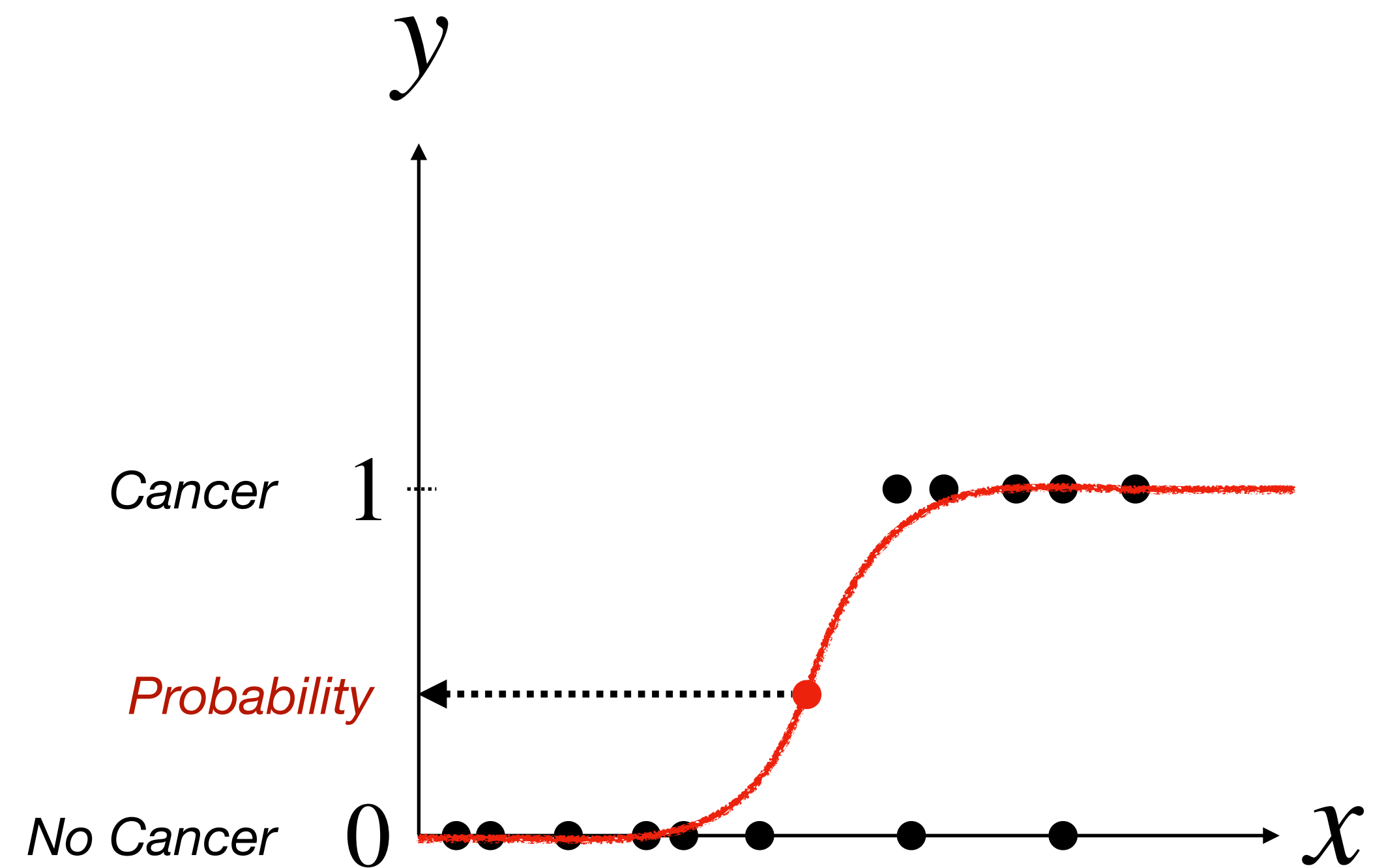
$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta^{\top}x)}} \quad \text{For } x \in \mathbb{R}^n$$

Where $\theta^{\top}x = \theta_0 + \theta_1x_1 + \theta_2x_2 + \dots$

$$\theta = [\theta_0, \theta_1, \dots]$$

$$x = [x_0, x_1, \dots]$$

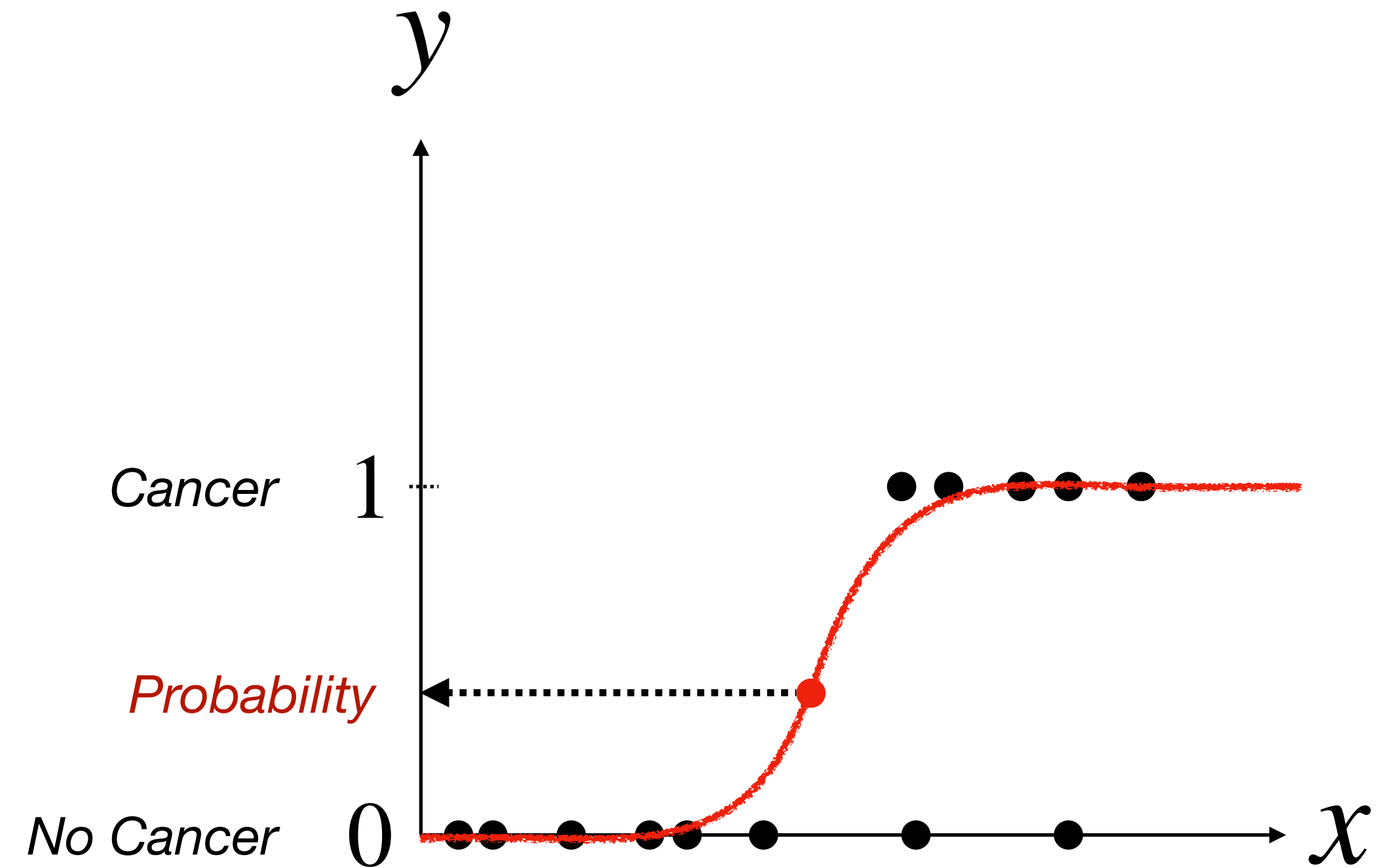
A **smooth** function that returns **probability of occurrence**



What if y is a label?

$$y = h_{\theta}(x) \quad \& \quad y \in [0,1]$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta^{\top}x)}}$$



1. **Define a predictor:** the logistic function ✓
2. **Define a loss:** distance between function and data ?
3. **Optimize loss**
4. **Test model**

How do we pick the best parameters θ ?

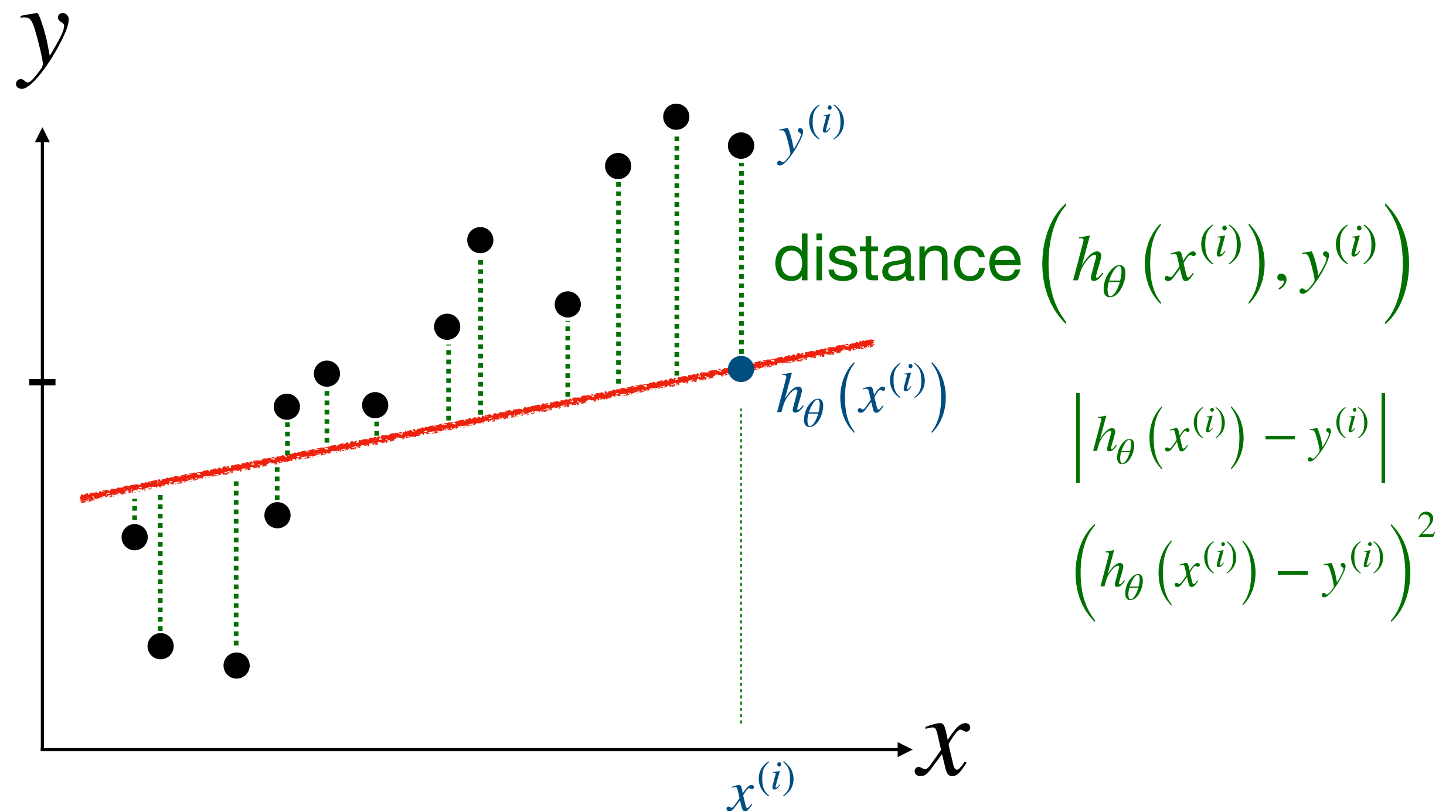
$$h_{\theta}(\mathbf{x}) = \theta^{\top} \mathbf{x} = \sum_{i=0}^d \theta_i x_i$$

Cost function

$$J(\theta) = \frac{1}{2} \sum_{i=1}^d \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

$$= \frac{1}{2} \sum_{i=1}^d \left(\theta^{\top} \mathbf{x}^{(i)} - y^{(i)} \right)^2$$

Ordinary least squares



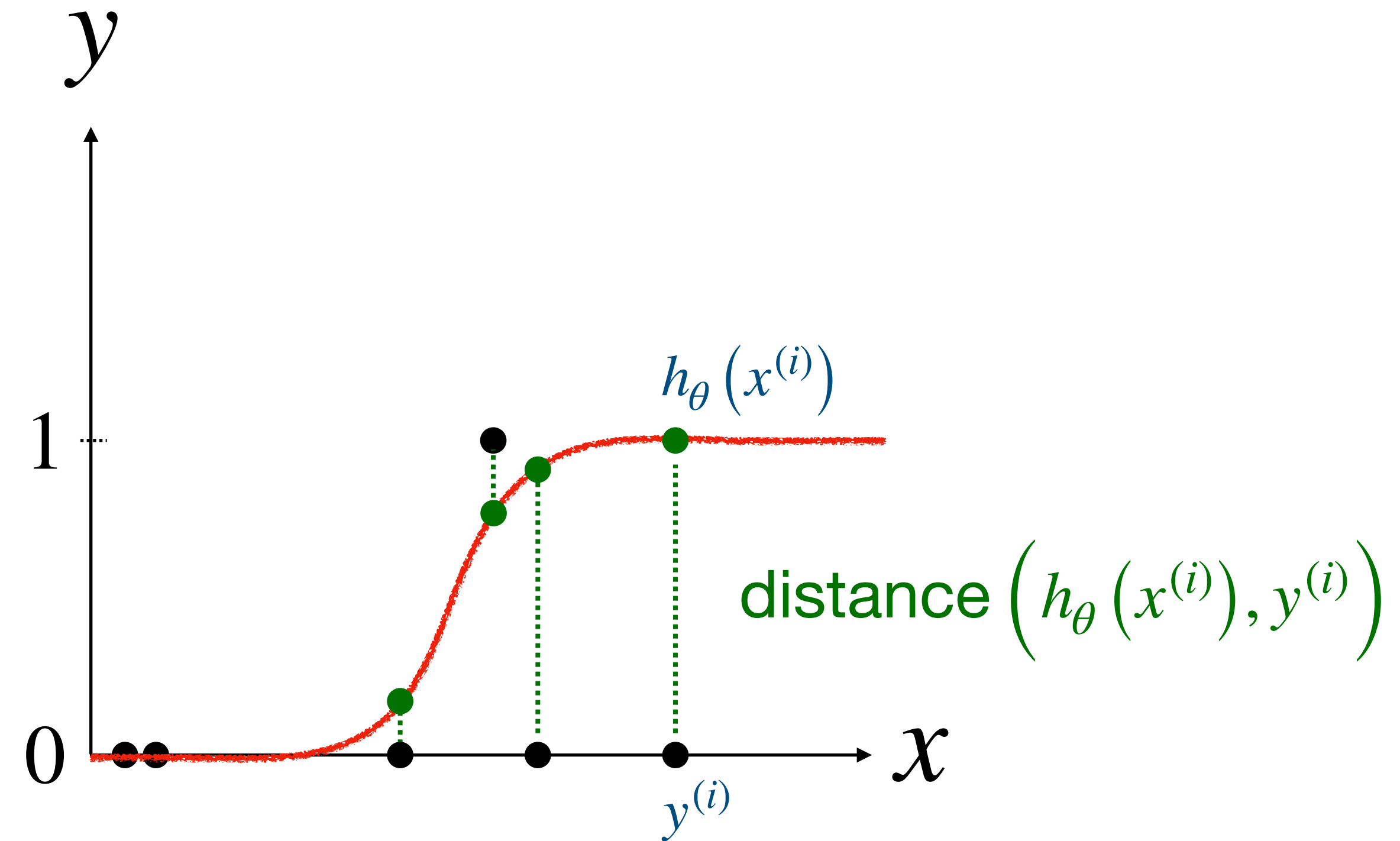
Logistic Regression

$$y = h_{\theta}(x)$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta^{\top} x)}} = \sigma(\theta^{\top} x)$$

Linear predictor
negative log-likelihood or OLS

$$J(\theta) = \frac{1}{2} \sum_{i=1}^d \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$



Logistic predictor
Binary-cross entropy loss

$$\mathcal{L}(\theta) = \sum_{i=1}^n y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)}))$$

Compute gradient $\nabla \mathcal{L}(\theta)$

Gradient descent $\rightarrow$ Done!

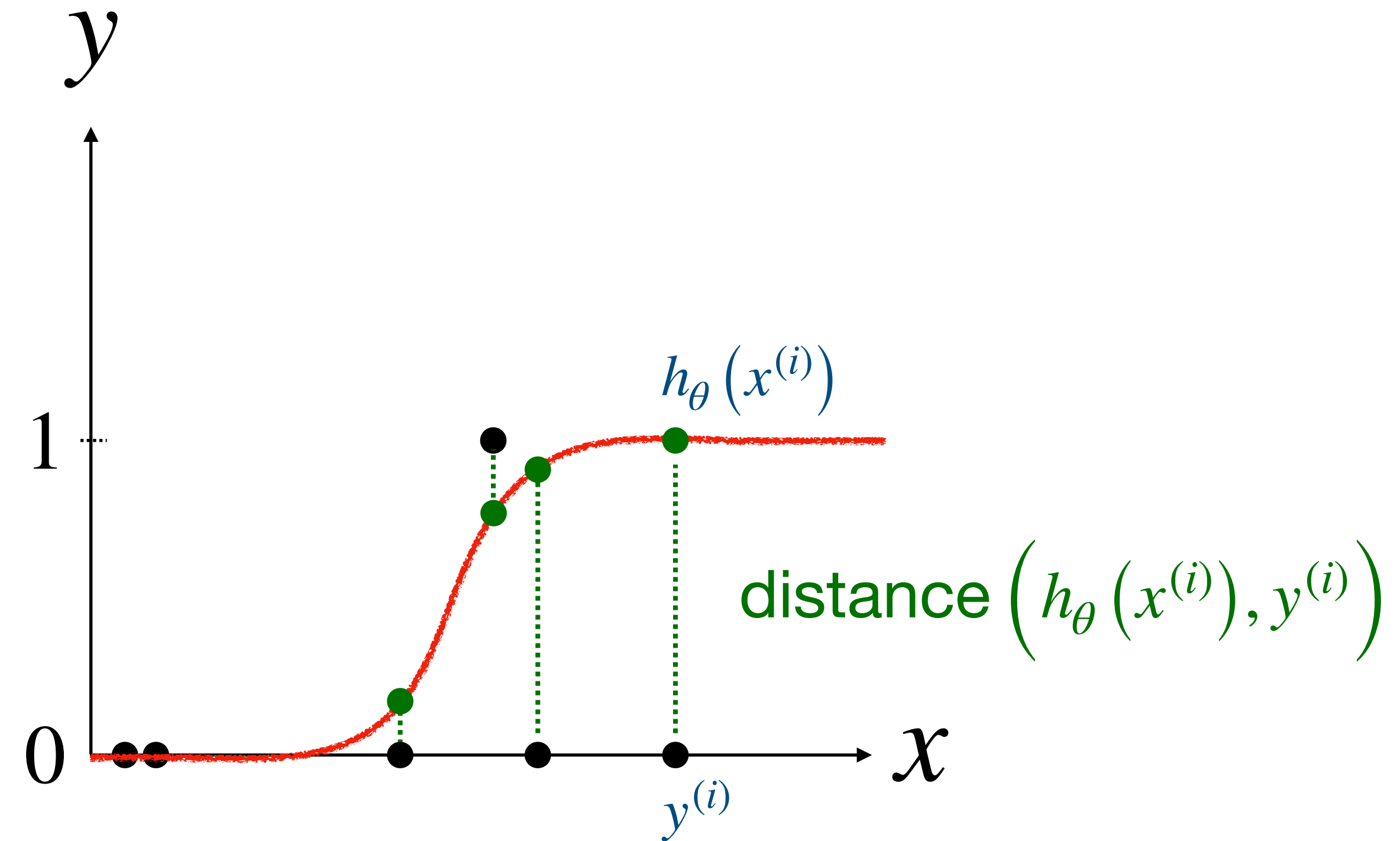
Logistic Regression

$$y = h_{\theta}(x)$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta^{\top} x)}} = \sigma(\theta^{\top} x)$$

Linear predictor
negative log-likelihood or OLS

$$J(\theta) = \frac{1}{2} \sum_{i=1}^d \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$



Why not use an ordinary least squares loss?

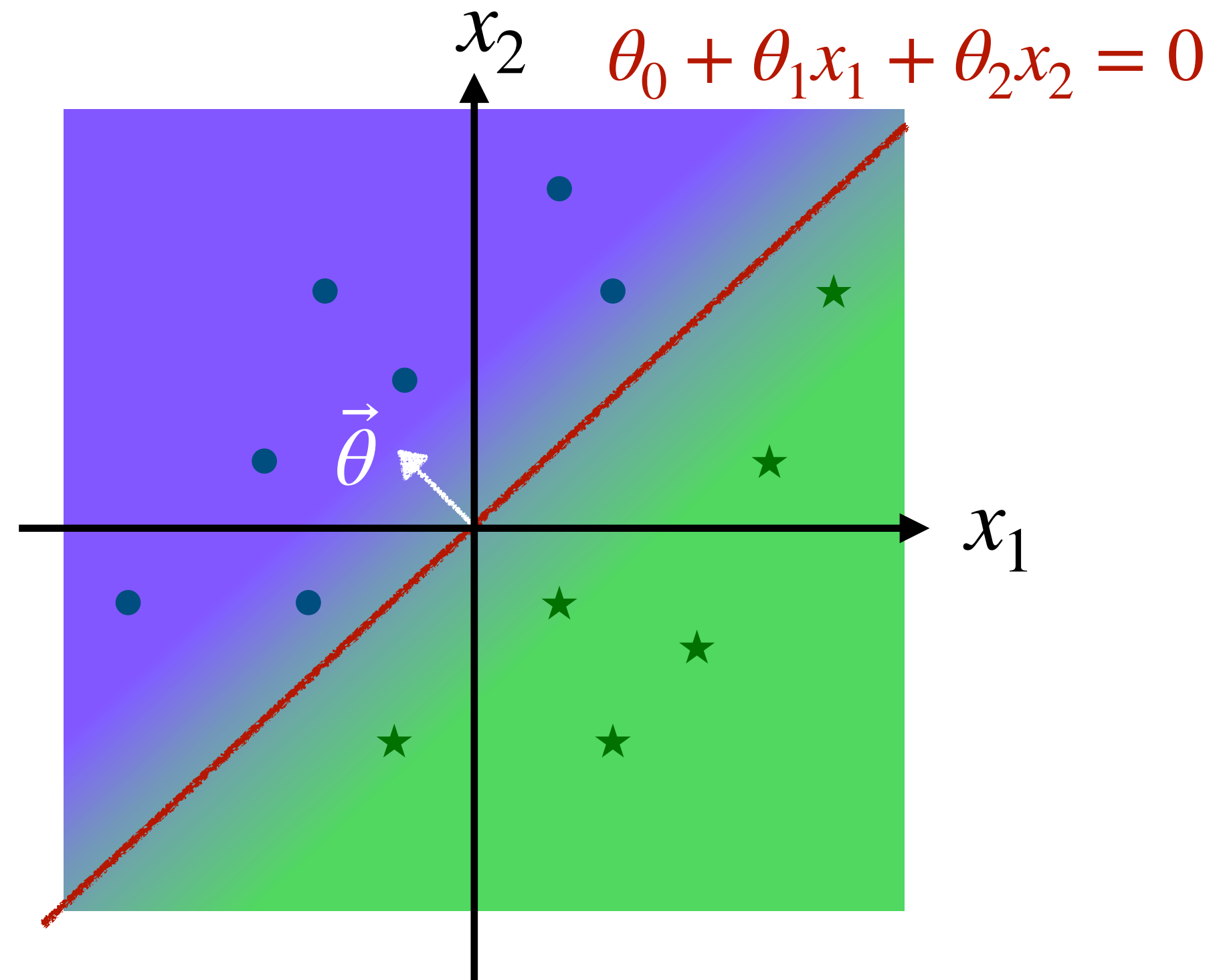
Higher dimensional inputs

For Logistic Regression and beyond

x_1	x_2	y
-2	-1	•
3	1	★
2	3	•
1	-1	★

⋮

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}}$$



how confident?

score

$$\theta^T x$$

For $y \in [1, -1]$

how correct?

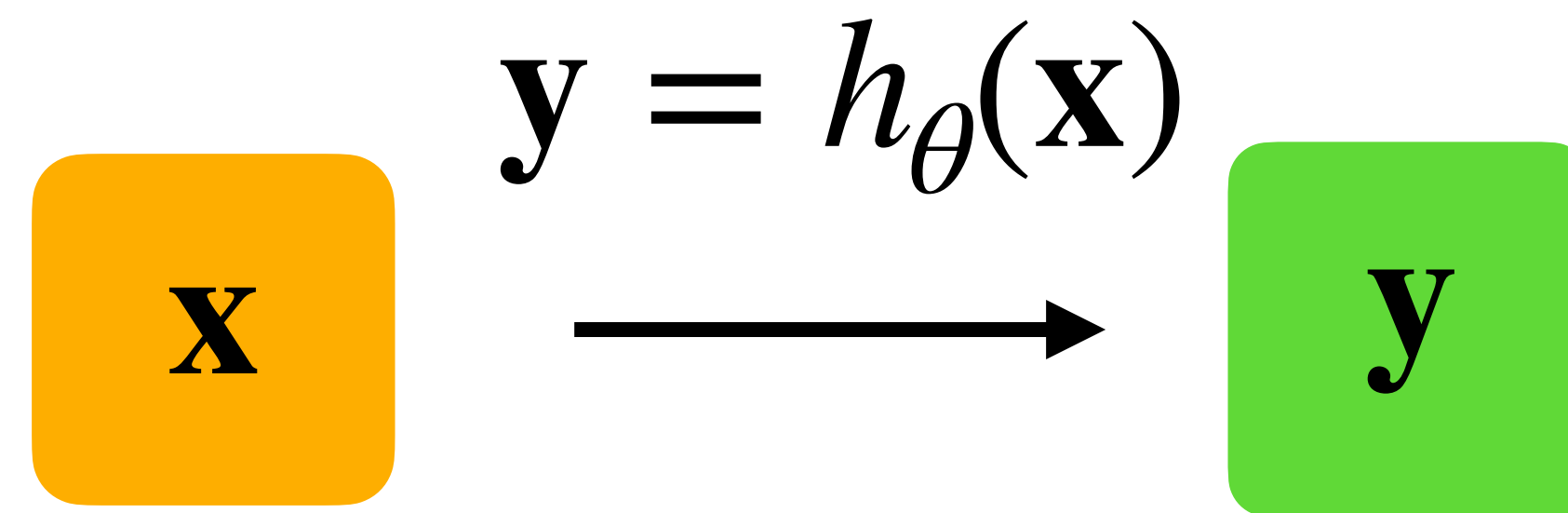
margin

$$(\theta^T x)y$$



Higher dimensional outputs

Multi-class classification - **Softmax** regression



k discrete values for representing output $\mathbf{y}$


$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

car

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

plane

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

boat

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

horse

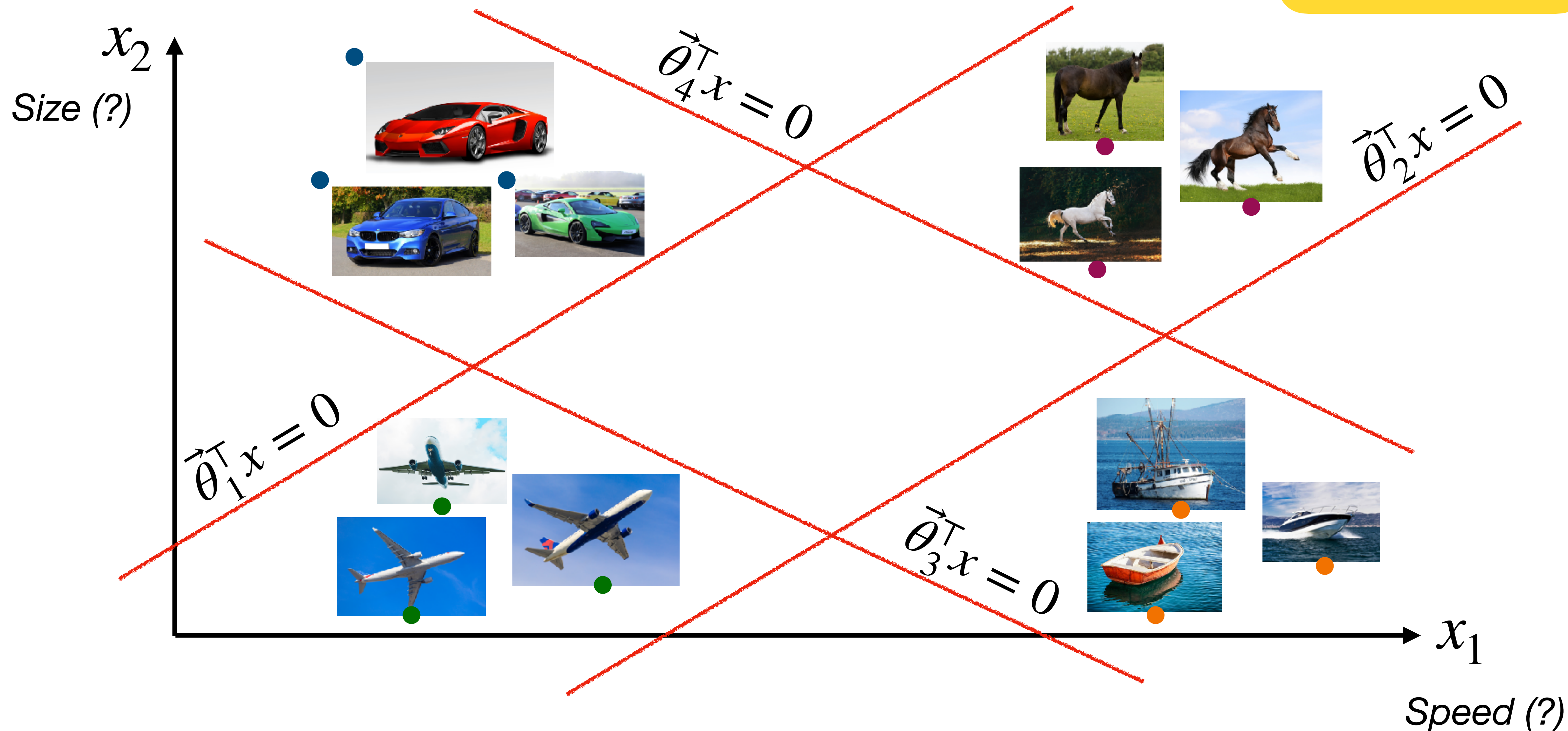
One-hot encoding

Multi-class classification - **Softmax**

WARNING!!!

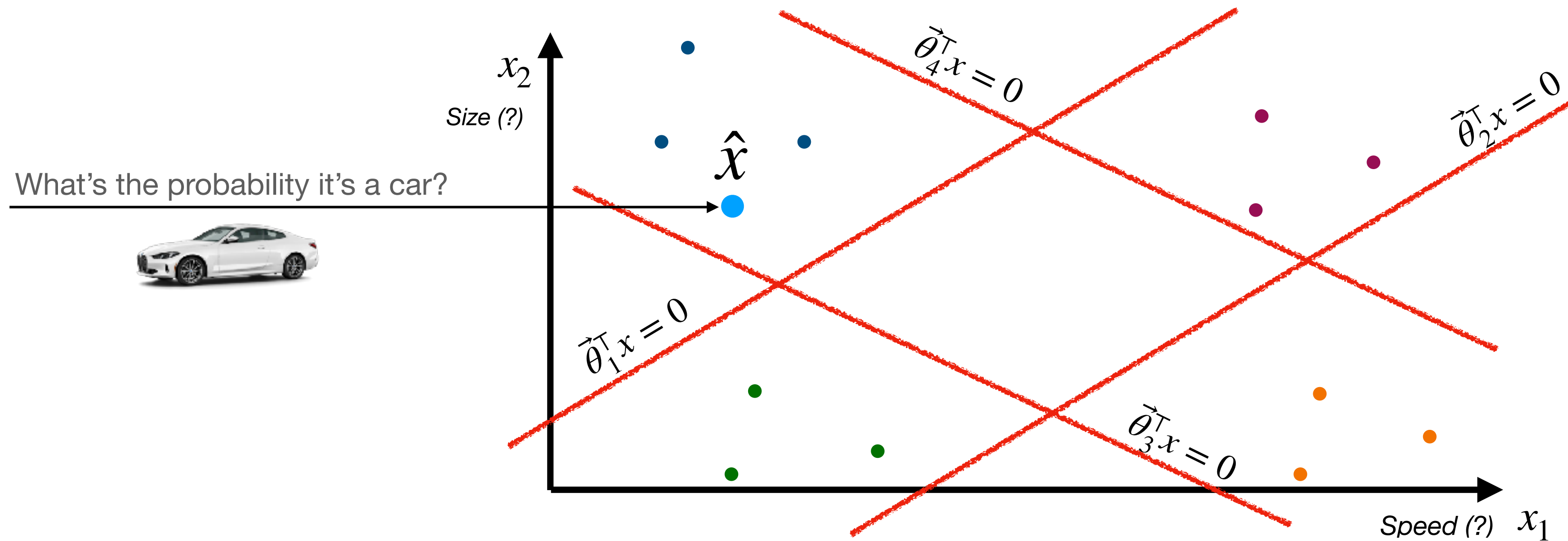
Notation Alert

$\vec{\theta}_i$ is a vector



How to turn **scores** into **probabilities**?

Note: This slides explains the intuition. Softmax regression is not 'exactly' equivalent to a combination of one-vs-all classification models



WARNING!!!

Notation Alert

$\vec{\theta}_i$ is a vector

Score

$$\vec{\theta}_1^T \hat{x} = 3$$

$$\vec{\theta}_2^T \hat{x} = -0.3$$

$$\vec{\theta}_3^T \hat{x} = -0.8$$

$$\vec{\theta}_4^T \hat{x} = -22$$

exp

Positive Measure

$$\exp(3) = 20.1$$

$$\exp(-0.3) = 0.75$$

$$\exp(-0.8) = 0.2$$

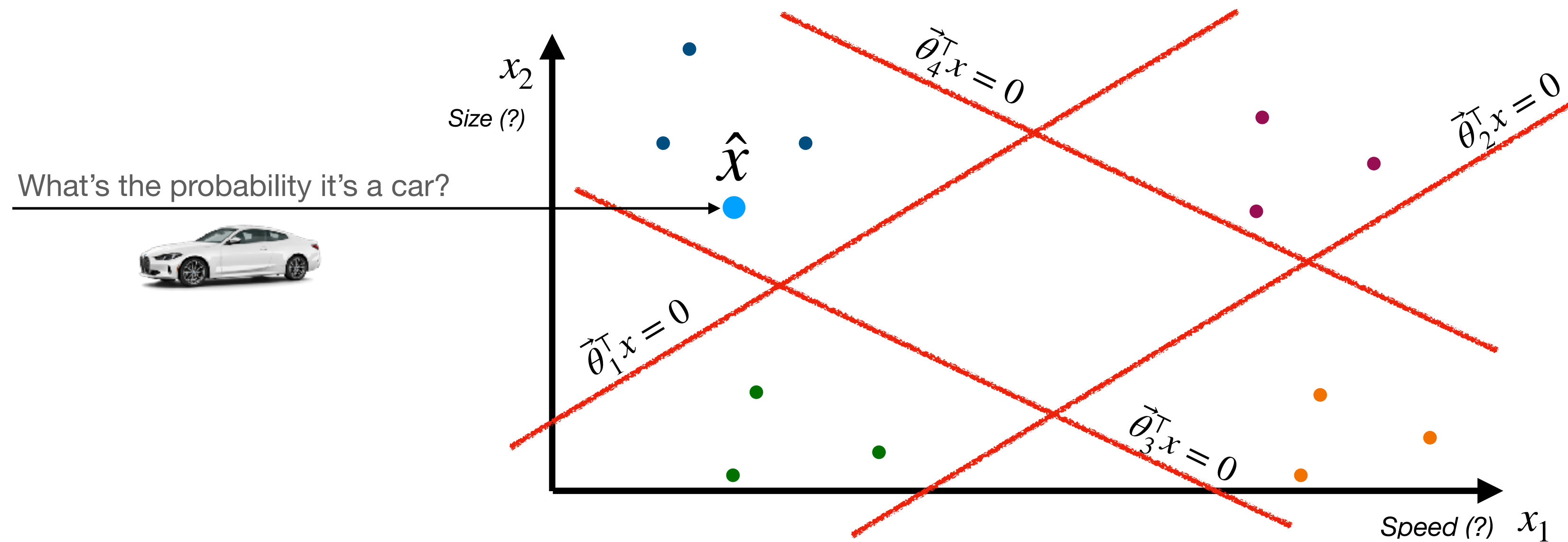
$$\exp(-22) = 0.00..1$$

Normalize

Softmax

$$\hat{p}(y = i | x; \theta) = \frac{\exp(\vec{\theta}_i^T x)}{\sum_{j=1}^k \exp(\vec{\theta}_j^T x)}$$

How to turn **scores** into **probabilities**?



WARNING!!!
Notation Alert
 $\vec{\theta}_i$ is a vector

Softmax

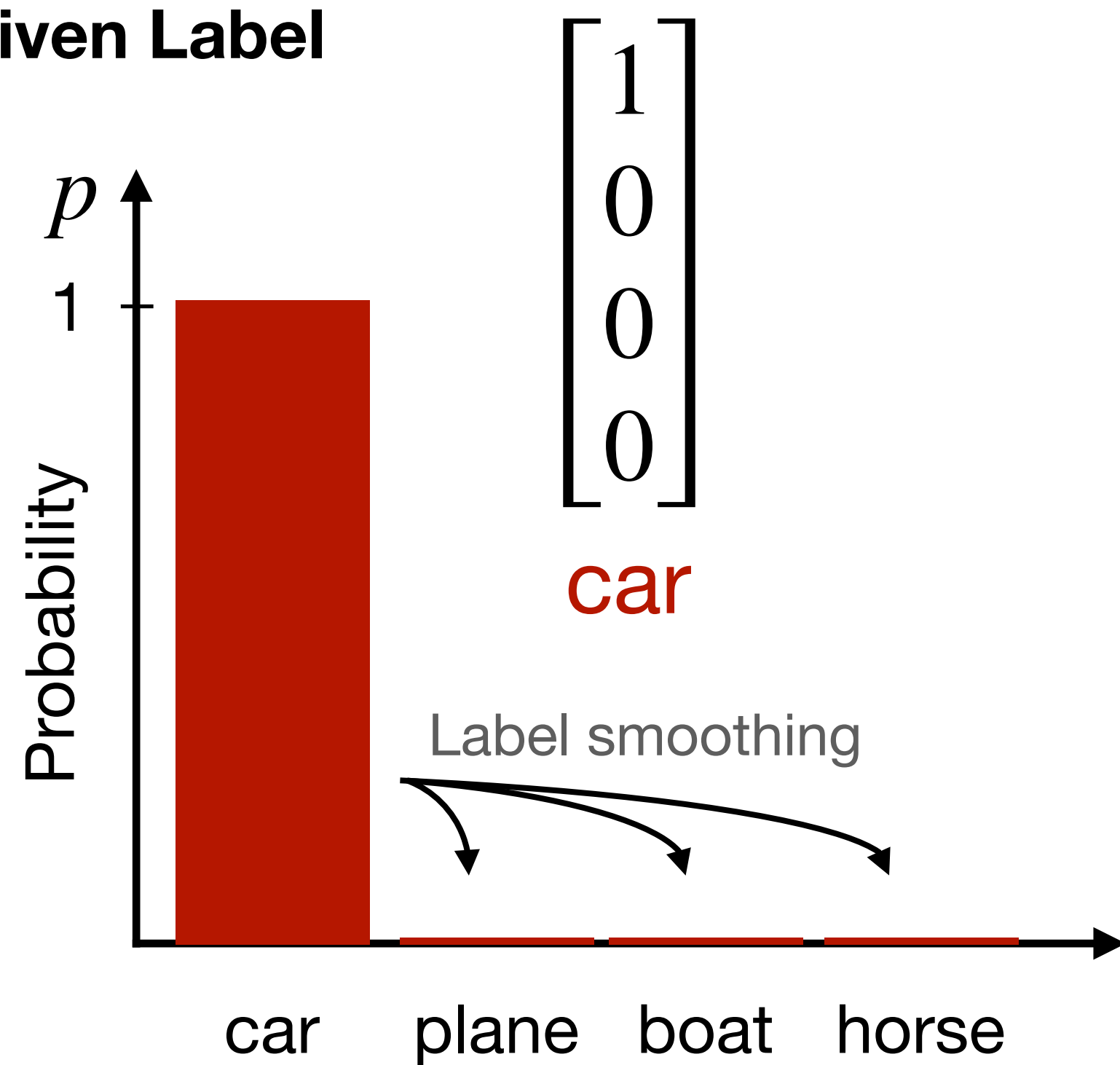
$$\hat{p}(y = i | x; \theta) = \frac{\exp(\vec{\theta}_i^T x)}{\sum_{j=1}^k \exp(\vec{\theta}_j^T x)}$$

$$\hat{p}_{\Theta} = \begin{bmatrix} \hat{p}(y = \text{car} | x; \theta_1) \\ \hat{p}(y = \text{plane} | x; \theta_2) \\ \hat{p}(y = \text{boat} | x; \theta_3) \\ \hat{p}(y = \text{horse} | x; \theta_4) \end{bmatrix}$$

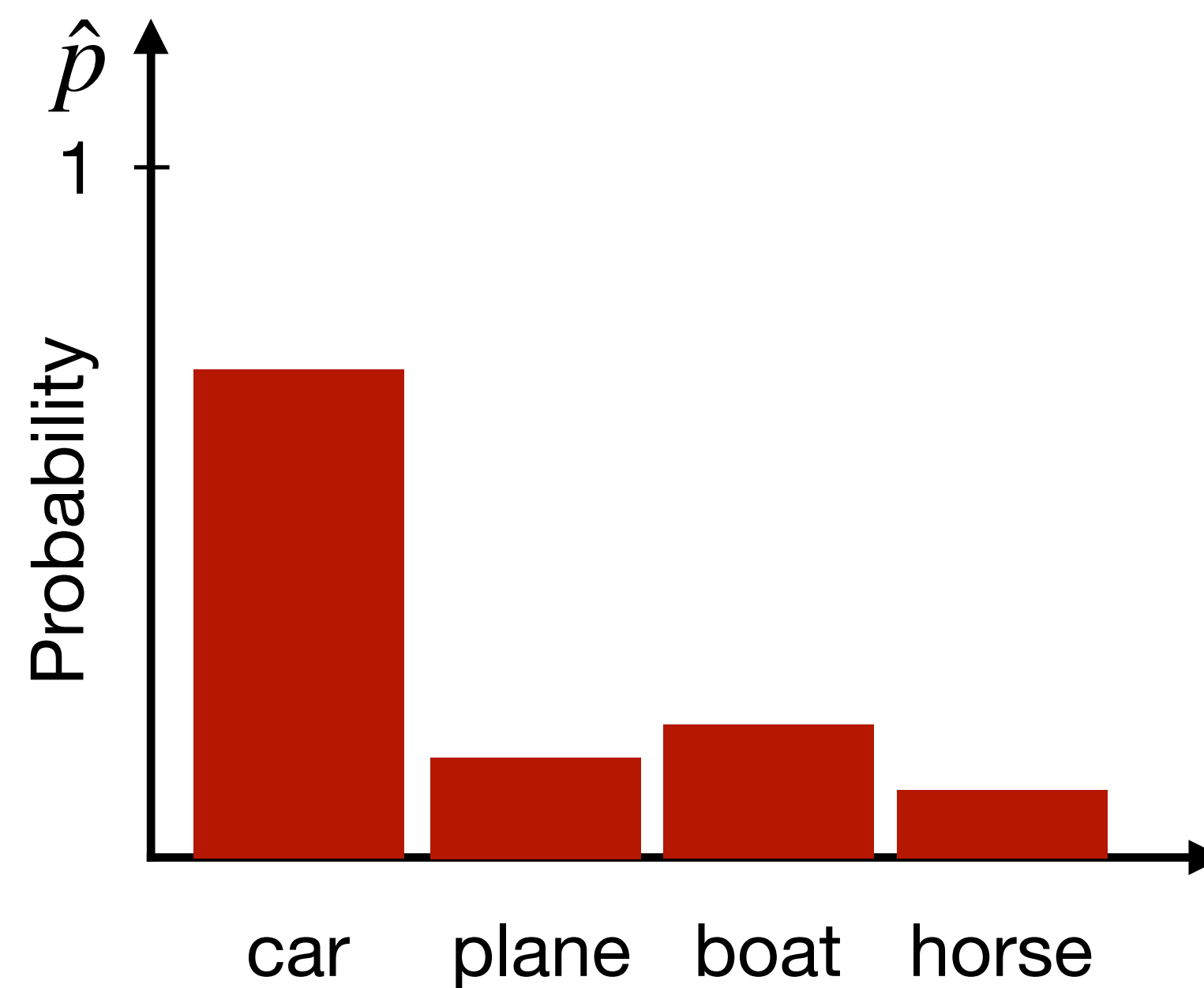
$$\hat{p}_{\Theta} = \text{Softmax}(\Theta x)$$

How do you train?

Given Label



Prediction



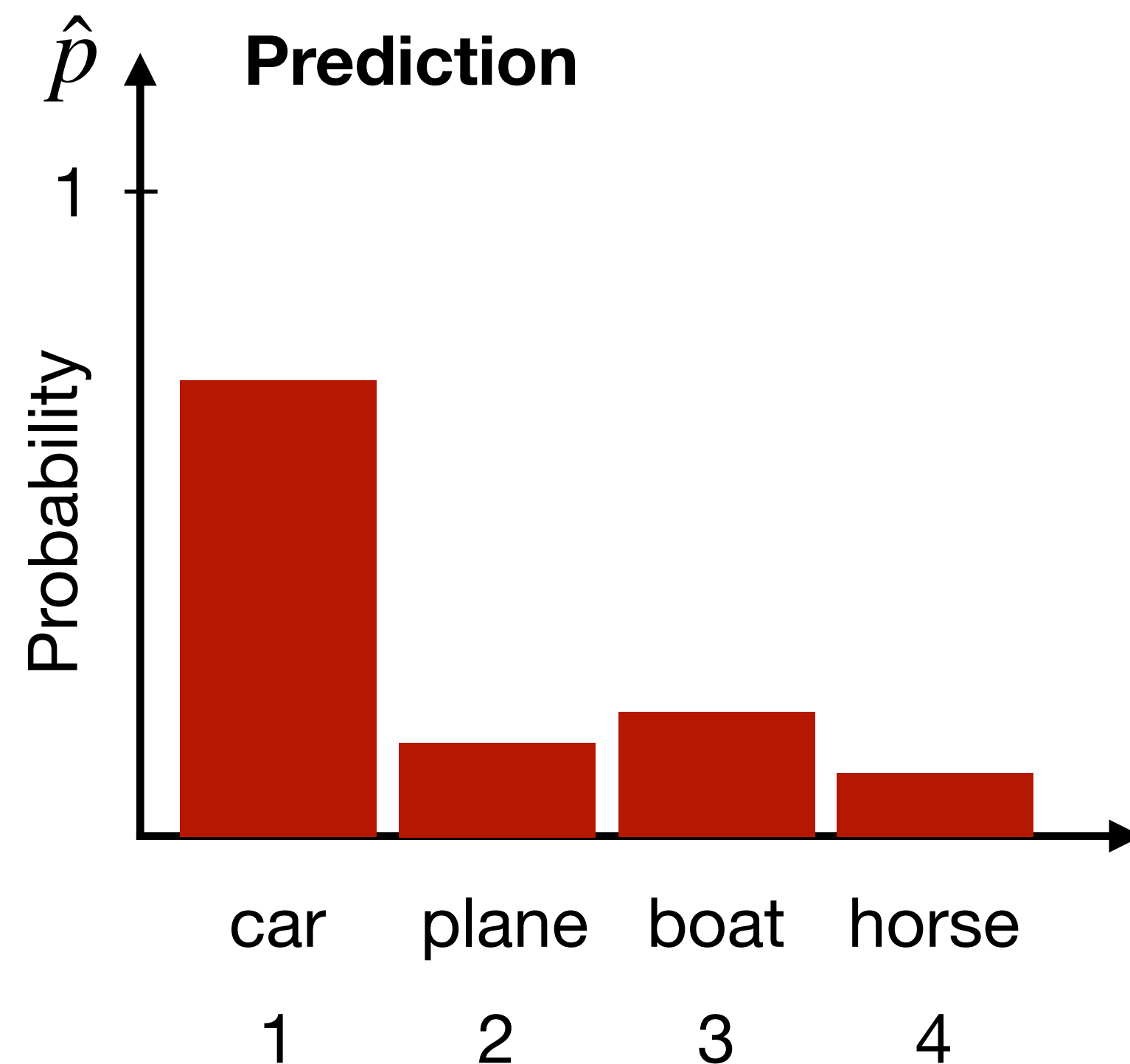
argmin

$\vec{\theta}_1, \vec{\theta}_2, \vec{\theta}_3, \vec{\theta}_4$

$$\begin{aligned} \text{CrossEntropy}(p, \hat{p}) &= - \sum_{i=1}^k p(y = i) \log(\hat{p}(y = i)) \\ &= - \log(\hat{p}(y = 1)) \end{aligned}$$

WARNING!!!
 y is the
ground-truth
index

How do you train?



$$\text{CrossEntropy}(p, \hat{p}) = - \sum_{i=1}^k p(y = i) \log (\hat{p}(y = i))$$

$$\text{Logit} = - \log (\hat{p}(y = 1))$$

$$= - \log \left(\frac{\exp(\vec{\theta}_i^\top x)}{\sum_{j=1}^k \exp(\vec{\theta}_j^\top x)} \right)$$

Train with Gradient Descent!